**E3.03**
**ISE3.3**

IMPERIAL COLLEGE LONDON

DEPARTMENT OF ELECTRICAL AND ELECTRONIC ENGINEERING
EXAMINATIONS 2010

EEE/ISE PART III/IV: MEng, BEng and ACGI

# COMMUNICATION SYSTEMS

Monday, 26 April 2:30 pm

Time allowed: 3:00 hours

**There are 19 questions on this paper.**

**Answer questions 1-10 and SIX other questions.**

*All answers should be given on the paper itself.*

*The compulsory questions 1-10 are multiple choice and together account for 40% of the marks. Each optional question accounts for 10% of the marks.*

*The following are provided:*
*A table of Fourier transforms*
*A Gaussian Tail Function graph*

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Examiners responsible:

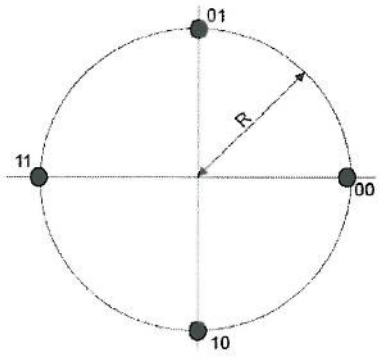
First Marker(s): A. Manikas
Second Marker(s): C. Ling

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1. If $D$ denotes the dimensionality of the signal space for an $M$-ary communication system which of the following statements is correct? [4 marks]

   (a) $D = 1$ for a binary ASK
   (b) $D = 2$ for a binary PSK
   (c) $D = 4$ for an $M$-ASK with $M = 4$
   (d) $D = 4$ for a QPSK
   (e) None of the above.

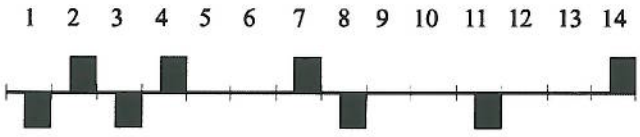
2. The constellation diagram of a QPSK communication system is shown below




   If the minimum distance is $2 \times \sqrt{2} \times 10^{-3}$ and the bit rate is 40kbits/sec what is the amplitude of this $M$-ary set of signals? [4 marks]

   (a) $\sqrt{8}$mV
   (b) 0.16V
   (c) 0.4 V
   (d) 0.8 V
   (e) None of the above.

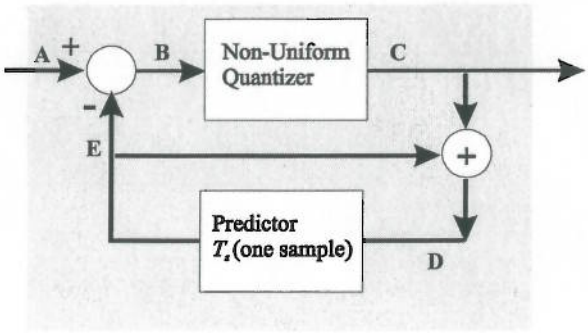
3. The following HDB3 encoded signal




   represents the binary sequence: [4 marks]

   (a) 0 1 1 1 0 0 0 1 0 0 0 0 0 1,
   (b) 1 1 1 1 0 0 1 1 0 0 1 0 0 1,
   (c) 1 1 1 0 0 0 0 0 0 0 0 0 0 1,
   (d) 0 1 1 0 0 0 0 1 0 0 0 0 0 1,
   (e) none of the above.

4. Consider the *mse*-differential quantizer shown in the following figure




which employs a 6-level non-uniform quantizer, having the following input and output levels [4 marks]

| I/P (volts) | O/P (volts) |
|---|---|
| $+7 <$input$\leq +255$ | $+25$ |
| $+3 <$input$\leq +7$ | $+5$ |
| $-1 <$input$\leq +3$ | $+2$ |
| $-4 <$input$\leq -1$ | $-2$ |
| $-8 <$input$\leq -4$ | $-5$ |
| $-255 \leq$input$\leq -8$ | $-25$ |

If the input is a step signal of amplitude 0V $\rightarrow$ 49V, and assuming "E"=0 as an initial value, then the data sequence at point 'D' is

(a) 25, 50, 48, 48, 50, 48, 50
(b) 25, 48, 50, 50, 48, 50, 48
(c) 25, 48, 50, 48, 50, 48, 50
(d) 25, 50, 48, 50, 48, 50, 48
(e) none of the above.

5. A digital communication system having an energy utilisation efficiency (EUE) equal to 40, operates in the presence of additive white Gaussian noise of double-sided power spectral density $\text{PSD}_n(f)=0.5\times10^{-6}$ W/Hz. If the channel capacity C is 16 kbits/s and the channel bandwidth B is 4 kHz, then the data bit rate $r_b$ is [4 marks]

(a) 1.5 kbits/sec,
(b) 2 kbits/sec,
(c) 3.5 kbits/sec,
(d) 4 kbits/sec,
(e) none of the above.

6. Consider that a white Gaussian noise-signal with power spectral density

$$\text{PSD}_{n_i}(f) = 10^{-6} W/Hz$$

is applied at the input of a filter with impulse response

$$h(t) = 8 \times 10^3 \sin c\{4 \times 10^3 t\} . \cos(2\pi 10^4 t)$$

The power spectral desity $\text{PSD}_n(f)$ of the signal at the output of the filter is [4 marks]

(a) $\text{PSD}_n(f) = 10^{-6}\left(\text{rect}\frac{f-4\times10^3}{10^4} + \text{rect}\frac{f+4\times10^3}{10^4}\right)$

(b) $\text{PSD}_n(f) = 10^{-6}\left(\text{rect}\frac{f-10^4}{4\times10^3} + \text{rect}\frac{f+10^4}{4\times10^3}\right)$

(c) $\text{PSD}_n(f) = 10^{-3}\left(\text{rect}\frac{f-4\times10^3}{10^4} + \text{rect}\frac{f+4\times10^3}{10^4}\right)$

(d) $\text{PSD}_n(f) = 10^{-3}\left(\text{rect}\frac{f-10^4}{4\times10^3} + \text{rect}\frac{f+10^4}{4\times10^3}\right)$

(e) none of the above.

7. With reference to Question 6, the power $P_n$ of the signal at the output of the filter is [4 marks]

(a) $20mW$

(b) $20W$

(c) $8mW$

(d) $8W$

(e) none of the above.

8. A 'short-code' BPSK DS/SSS uses an m-sequence and a data rate 9.6 kbits/sec. If it is required that the spread spectrum signal will have bandwidth no larger than 25MHz, what is the largest period of the m-sequence that can be used? [4 marks]

(a) 255

(b) 511

(c) 1023

(d) 2047

(e) None of the above

9. A speech signal having a maximum frequency of 4kHz is sampled at twice the Nyquist rate and then fed through an 8-bit uniform quantizer. The generated binary sequence is then fed through a binary PSK direct sequence spread spectrum system which operates in the presence of a jammer of power 1.6 Watts and in the presence of additive white Gaussian noise with double-sided power spectral density $0.5 \times 10^{-12}$ Watts/Hz. The amplitude of the BPSK signal is 0.5V.

   For this system, the spread spectrum bandwidth $B_{ss}$ is 32 MHz and the system is not synchronised.

   The power of the noise at the output of the despreader is [4 marks]

   (a) 16 $\mu$W
   (b) 32 $\mu$W
   (c) 0.064 $\mu$W
   (d) 0.128 $\mu$W
   (e) none of the above.

10. With reference to Question 9, the power of the jammer at the output of the despreader is [4 marks]

   (a) 1.6 $m$W
   (b) 3.2 $m$W
   (c) 6.4 $m$W
   (d) 12.8 $m$W
   (e) none of the above

11. Consider a PCM system where its sampler operates at twice the Nyquist rate and its quantizer consists of an $\mu$-law compander (with $\mu = 100$) followed by a $Q$-level uniform quantizer.
Note that $\mu$-law compression is defined as follows:

$$\text{output} = \frac{\ln\left(1+\mu.\left|\frac{x}{x_{\max}}\right|\right)}{\ln(1+\mu)} x_{\max} \times \text{sign}(x) \qquad \text{where} \begin{cases} x = \text{input value in } Volts \\ x_{\max} = \text{maximum input value in } Volts \\ \text{sign}(x) = \text{ the sign of } x \end{cases}$$

The maximum and minimum values of the input signal are +5Volts and −5Volts respectively and it occupies a 15kHz bandwidth. It is required that the Signal-to-Quantization-Noise Ratio (SNRq) should be better than 8.5 dB

(a) Find the minimum number of quantizing levels $Q$ required, as well as the transmission bit rate. [4 marks]

(b) If the signal at the output of the sampler at time $kT_s$ is equal to −2.4 Volts, estimate the instantanous quantization noise $n_q(kT_s)$ where $T_s$ is the sampling period and $k$ is an integer. [6 marks]

12. Consider two analogue signals x($t$) and y($t$). The signal x($t$) has a uniform amplitude probability density function, i.e.

$$\text{pdf}_{\text{x}}(x) = \frac{1}{3}\text{rect}\left\{\frac{x}{3}\right\}$$

while the signal y(t) has a Gaussian amplitude probability density function $\text{pdf}_{\text{y}}(y)$ of mean $\mu_y$ and standard deviation equal to $\sqrt{P_x}$where $P_x$ denotes the average power of x($t$).

Find

(a) the average power $P_x$ of the signal x($t$). [2 marks]

(b) the differential entropy H(x) of the signal x($t$) [2 marks]

(c) the differential entropy H(y) of y($t$) [2 marks]

(d) the entropy power of x(t) [2 marks]

(e) the entropy power of y(t) [2 marks]

13. Consider a random binary sequence of 0's and 1's, which are statistically independent with Pr(1) = Pr(0) = 0.5. This binary sequence is transmitted as a random signal with 1's and 0's being sent using the pulses $s_1(t)$ and $s_0(t)$ described below:

$$0 \longmapsto s_0(t) = 3 \times 10^{-3} \times \Lambda\left\{\frac{t}{1ms}\right\} \text{ Volts}$$

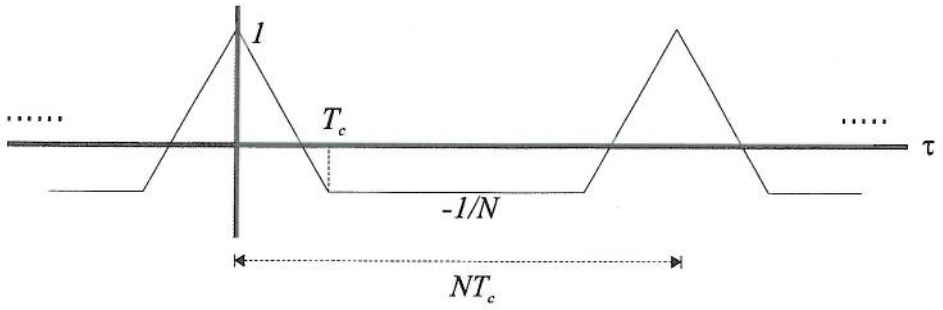
and

$$1 \longmapsto s_1(t) = -3 \times 10^{-3} \times \Lambda\left\{\frac{t}{1ms}\right\} \text{ Volts}$$

(a) Find the Power Spectral Density of the transmitted signal. [6 marks]

(b) What is the dimensionality $D$ of the signal space? Plot the constellation diagram. [4 marks]

14. The autocorrelation function $R_{bb}(\tau)$ of a pseudo-random signal (m-sequence) $b(t)$ of a spread spectrum communication system is given below:




(a) Write a mathematical expression, using Woodward's notation, to describe the above autocorrelation function. [5 marks]

(b) Find the power spectral density PSD($f$) of $b(t)$. [5 marks]

15. (a) Sketch the feedback shift register whose feedback connections are represented by the primitive polynomial $x^4 + x + 1$ and find the length $N$ of this sequence. [4 marks]

(b) If the clock rate is 2.7 $Mb/sec$, find the period of this sequence. [6 marks]

16. A speech signal having a maximum frequency of 4kHz is sampled at twice the Nyquist rate and then fed through an 8-bit uniform quantizer. The generated binary sequence is then fed through a binary PSK direct sequence spread spectrum system which operates in the presence of a jammer of power 1.6 Watts and in the presence of additive white Gaussian noise with double-sided power spectral density $0.5 \times 10^{-12}$ Watts/Hz. The amplitude of the BPSK signal is 0.5V.

For this system, the spread spectrum bandwidth $B_{ss}$ is 32 MHz and the system is not synchronised.

Find:

(a) the power of the noise at the output of the correlator, [5 marks]

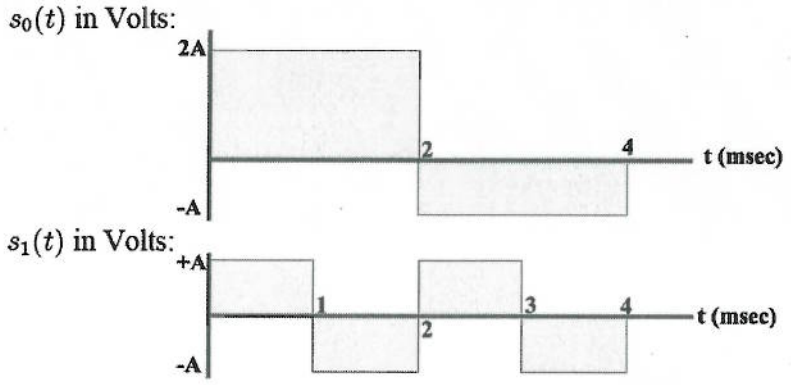
(b) the power of the jammer at the output of the correlator. [5 marks]

17. (a) A high quality music signal with a Crest Factor of 4.4668, having a maximum frequency 18 kHz, is applied to a uniform PCM system (i.e. PCM with a uniform quantizer). If it is specified that the Signal-to-Quantisation-Noise ratio ($SNR_q$) should be better than 50dB, find the minimum data bit rate required [6 marks]

(b) In a binary PCM communication system prove that bandwidth expansion factor $\beta$ is equal to the average number of bits $\gamma$ per quantization level, i.e. [4 marks]

$$\beta = \gamma$$

18. Given a binary digital communication system which employs the following two equally-probable signals and an optimum digital demodulator (matched filter or correlation receiver)




The received signals are corrupted by additive white Gaussian channel noise having a double-sided power spectral density of $\frac{7}{8}10^{-9}$ W/Hz. Find

(a) the energy utilisation efficiency EUE as a function of the amplitude $A$; [5 marks]

(b) the amplitude $A$ if the bit-error rate is $2.2 \times 10^{-2}$. [5 marks]

19. Consider a binary pulse-code-modulation (binary-PCM) system in which an ideal sampler and a uniform $Q$-level quantiser is used, and the output average Signal-to-Noise power ratio ($\text{SNR}_{\text{out}}$) of the receiver is given by:

$$\text{SNR}_{\text{out}} = \frac{2^{2\gamma}}{1 + 4p_e 2^{2\gamma}}$$

where $p_e$ depends on the modulation scheme being used, and $\gamma$ is the number of bits per quantisation level.

(a) Derive an expression for the probability, $p_e$, of a bit in error when threshold effects occur. [5 marks]

(b) Given that $Q = 8$ and that the modulation scheme being used is phase-shift-key (PSK), i.e.

$$\begin{aligned} s_0(t) &= A.\sin(2\pi F_c t) \text{ for } 0 \le t \le T_{cs} \\ s_1(t) &= A.\sin(2\pi F_c t + \pi) \text{ for } 0 \le t \le T_{cs} \end{aligned}$$

with $F_c = \frac{5}{T_{cs}}$ where $T_{cs}$ =bit duration= 4 $\mu$s, find the amplitude $A$ at which threshold occurs. The channel noise is assumed white Gaussian of double-sided power spectral density $4 \times 10^{-12}$W/Hz [5 marks]

## *FOURIER TRANSFORMS - TABLES*

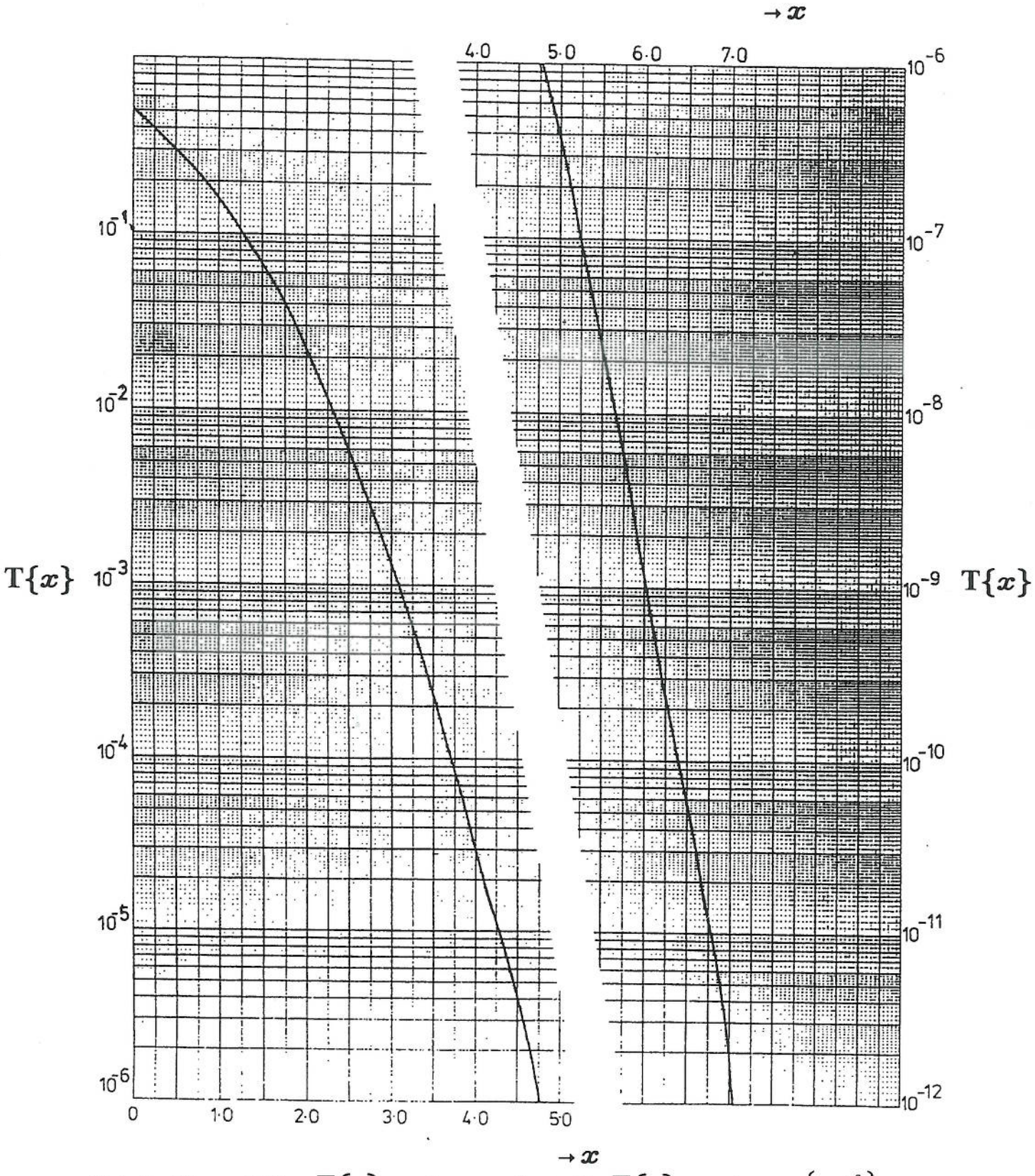
| | DESCRIPTION | FUNCTION | TRANSFORM |
|---|---|---|---|
| 1 | Definition | $g(t)$ | $G(f) = \int_{-\infty}^{\infty} g(t).e^{-j2\pi ft}dt$ |
| 2 | Scaling | $g(\frac{t}{T})$ | $\|T\| \, . \, G(fT)$ |
| 3 | Time shift | $g(t-T)$ | $G(f) \, . \, e^{-j2\pi fT}$ |
| 4 | Frequency shift | $g(t) \, . \, e^{j2\pi Ft}$ | $G(f-F)$ |
| 5 | Complex conjugate | $g^*(t)$ | $G^*(-f)$ |
| 6 | Temporal derivative | $\frac{d^n}{dt^n} \, . \, g(t)$ | $(j2\pi f)^n . \, G(f)$ |
| 7 | Spectral derivative | $(-j2\pi t)^n . g(t)$ | $\frac{d^n}{df^n} \, . \, G(f)$ |
| 8 | Reciprocity | $G(t)$ | $g(-f)$ |
| 9 | Linearity | $A \, . \, g(t) + B \, . \, h(t)$ | $A \, . \, G(f) + B \, . \, H(f)$ |
| 10 | Multiplication | $g(t) \, . \, h(t)$ | $G(f) * H(f)$ |
| 11 | Convolution | $g(t) * h(t)$ | $G(f) \, . \, H(f)$ |
| 12 | Delta function | $\delta(t)$ | 1 |
| 13 | Constant | 1 | $\delta(f)$ |

| | DESCRIPTION | FUNCTION | TRANSFORM |
|---|---|---|---|
| 14 | Rectangular function | $\mathbf{rect}\{t\} \equiv \begin{cases} 1 & \text{if } \|t\| < \frac{1}{2} \\ 0 & otherwise \end{cases}$ | $\mathbf{sinc}(f) = \frac{\sin \pi f}{\pi f}$ |
| 15 | Sinc function | $\mathbf{sinc}(t)$ | $\mathbf{rect}(f)$ |
| 16 | Unit step function | $u(t) = \begin{cases} +1, & t > 0 \\ 0, & t < 0 \end{cases}$ | $\frac{1}{2}\delta(f) - \frac{j}{2\pi f}$ |
| 17 | Signum function | $\mathbf{sgn}(t) = \begin{cases} +1, & t > 0 \\ -1, & t < 0 \end{cases}$ | $-\frac{j}{\pi f}$ |
| 18 | Decaying exponential (two-sided) | $e^{-\|t\|}$ | $\frac{2}{1+(2\pi f)^2}$ |
| 19 | Decaying exponential (one-sided) | $e^{-\|t\|}.u(t)$ | $\frac{1-j2\pi f}{1+(2\pi f)^2}$ |
| 20 | Gaussian function | $e^{-\pi t^2}$ | $e^{-\pi f^2}$ |
| 21 | Lambda function | $\Lambda\{t\} \equiv \begin{cases} 1-t & \text{if } 0 \le t \le 1 \\ 1+t & \text{if } -1 \le t \le 0 \end{cases}$ | $\mathbf{sinc}^2(f)$ |
| 22 | Repeated function | $\mathbf{rep}_T\{g(t)\} = g(t) * \mathbf{rep}_T\{\delta(t)\}$ | $\|\frac{1}{T}\|.\mathbf{comb}_{\frac{1}{T}}\{G(f)\}$ |
| 23 | Sampled function | $\mathbf{comb}_T\{g(t)\} = g(t).\mathbf{rep}_T\{\delta(t)\}$ | $\|\frac{1}{T}\|.\mathbf{rep}_{\frac{1}{T}}\{G(f)\}$ |

# Tail Function Graph

The graph below shows the Tail function $\mathbf{T}\{x\}$ which represents the area from $x$ to $\infty$ of the Gaussian probability density function N(0,1), i.e.

$$\mathbf{T}\{x\} = \int_x^{\infty} \frac{1}{\sqrt{2\pi}} . \exp\left( - \frac{y^2}{2} \right) dy$$




Note that if $x > 6.5$ then $\mathbf{T}\{x\}$ may be approximated by $\mathbf{T}\{x\} \approx \frac{1}{\sqrt{2\pi} . x} . \exp\left\{ - \frac{x^2}{2} \right\}$